The Threshold Does Not Transfer: Open-Set Test-Time Adaptation Across Buildings

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ABSTRACT

Deploying an activity recogniser in a new building means facing two shifts at once: the sensors are in different places, and the occupant may do things the model was never trained on. Open-set and universal domain adaptation address the second, and test-time adaptation the first, but the methods for both are evaluated offline, with repeated passes over a target set that is also the evaluation set. We study the combined shift under constraints a deployment can meet: a source model trained once and frozen, a single online pass over the unlabelled target stream in recording order, no source data, and no target labels anywhere, including for the threshold that decides what counts as unknown.

Measured against a model trained on the target building at matched sample size, the cross-building penalty on known activities is 1.6 macro-F_1 points entering one apartment and 36.0 entering the other, a wider asymmetry than the 0.6 and 21.3 of the closed-set case [1]. Four test-time methods follow it: none improves on a frozen source model where there is no shift to recover. One appears to, and the appearance is instructive. PETAL records the best open-set H-score in one direction while classifying known activities roughly half as well, because degrading a model uniformly lowers its confidence everywhere, pushes more windows below a fixed threshold, and raises the unknown-detection term that H is bottlenecked by. The standard open-set score is therefore maximised in part by damage, and pairing it with known-class accuracy does not remove the problem; it only makes it visible.

The rejection rule fails for a different reason. Maximum softmax, entropy, maximum logit and free energy are all at or below chance as novelty detectors here, with bootstrap intervals excluding 0.5 for three of the four: the model is as confident on activities it never saw as on those it was trained on. A Mahalanobis distance to source class prototypes is the only score above chance, and at a matched rejection rate it catches 1.58 and 1.17 times as many unknowns as rejecting at random — real, and far smaller than the H-score comparison suggests, because two scores held at the same nominal percentile reject very different fractions of a target stream.

That last observation is the main result. A threshold set to reject 1% of source validation rejects between 7% and 63% of the target stream, depending only on which score carries it. Improving detection does not fix it: a source-only weighting rule that gains 0.064 AUROC in the direction with headroom *loses* 0.11 H at the threshold a deployment would use, while the variant with the best H in both directions has significantly worse AUROC in one. Detection quality and deployed open-set performance come apart, so cross-domain threshold calibration is the binding constraint rather than one difficulty among several. We recommend reporting the rejection rate, which is observable without labels, beside every open-set score.

1. Introduction

1.1. Motivation and protocol

Measured against a model trained on the target building at matched sample size, the cross-building penalty for activity recognition in this pair of apartments is 0.6 macro-F1 points entering S_2 and 21.3 entering S_1: in one direction there is essentially no distribution shift for an adaptation method to recover, and in the other there is a substantial one. That asymmetry, rather than any single method's score, is what decided the outcome of our closed-set study, and the methods tracked it — losing ground wherever there was no headroom and gaining only where there was.

This section asks whether a second, simultaneous shift changes that. We withhold two activities, `toileting` and `cleaning`, from source training and treat them as target-private, so the model faces a change of building and a change of label space at once. The retained classes are `meal_prep`, `watching_tv` and `music`; `cleaning` is placed in the unknown set deliberately, since at 113 target windows it is not learnable in this data and leaving it among the known classes would confound open-set performance with a sample-size limit. The experiment is a 2×2 grid, $\{S_1 \rightarrow S_2, S_2 \rightarrow S_1\}$ crossed with {closed-set, open-set}, and because withholding classes changes the source model, every cell is given its own penalty measurement before any method is run.

Throughout we hold to the Tier-1 definition of test-time adaptation used in the closed-set study: the source model is trained once and then frozen, with no further multi-epoch training on target data; adaptation is online, in a single pass over the unlabelled target stream, in recording order, with each batch predicted before the model is updated on it; source data is unavailable during adaptation; and no target labels are used anywhere, including for selecting hyperparameters and for calibrating the unknown-rejection threshold, which is fixed from source validation before the target stream is read. Any quantity a method relies on — scalers, prototypes, thresholds — must be estimable without target labels. Where we report a figure that violates this, as

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with the labelled ceiling of Section 2.6 and the matched-rejection-rate comparison of Section 2.4, it is labelled a diagnostic and is used to bound what a method could achieve rather than to claim what one did.

The constraint matters because the reference implementations in this area do not observe it. Section 2.8 documents four places where the released code for LEAD leaves the test-time setting, including fifty offline epochs over the target set and an evaluation loader that wraps the same data the model adapted on. Results obtained that way are not comparable with results obtained under the constraints above, which is why every number in this section is reported against our own Tier-1 source-only baseline rather than against published figures.

What to deploy

This section records, before the analysis that justifies it, the one configuration in this study that we would put in a building.

Frozen source model, no adaptation, Mahalanobis rejection.

Train once on the source building's known activities. Do not adapt at test time. Reject a window as unknown when its squared Mahalanobis distance to the nearest source class prototype exceeds the 95th percentile of that distance on the source validation split, with the prototypes and the threshold both fixed before the target stream is read.

H-score 0.5176 entering S_2, 0.4289 entering S_1.

Every component of this is Tier-1: no target labels, no target statistics, one pass, and no quantity estimated from the stream it is scored on. It is also, deliberately, a system with no adaptation in it. Sections 2.2 to 2.6 report what each of the parts we removed was worth, and none of them was worth keeping: the four adaptation methods cost between 12 and 27 H-score points, the confidence-based rejection scores are at or below chance, and the learned embedding contributes nothing the raw features do not.

We state the recommendation first because the rest of this section is a sequence of negative results, and a reader is entitled to know what we think survives them.

2. Results

2.1. The cross-building penalty in four cells

Withholding activities from source training changes the source model, so each cell of the grid needs its own baseline before any method is run. We reuse the measurement of the closed-set study [1]: hold the evaluation set fixed, vary only where the training labels came from, match training-set size by subsampling, and group the split by recording day. In the open-set cells the penalty is computed on known-class windows only, since a source model that never saw the withheld activities cannot be asked to classify them.

Two points about how these numbers should be read.

The closed-set rows reproduce 0.6 and 21.3 points exactly, on a harness written separately for this section. That is the only validation available to us that the measurement is implemented correctly rather than merely reported consistently.

Table 1
Cross-building penalty, known-class macro-F_1, matched training-set size, identical held-out target days. Closed-set rows reproduce [1] on an independently written harness.

Setting	Direction	Own days	Other building	Penalty
Closed	$S_1 \to S_2$	0.3497 ± 0.148	0.3434 ± 0.177	0.6
Closed	$S_2 \to S_1$	0.4897 ± 0.058	0.2767 ± 0.091	21.3
Open	$S_1 \to S_2$	0.8082 ± 0.133	0.7926 ± 0.137	1.6
Open	$S_2 \to S_1$	0.9153 ± 0.091	0.5557 ± 0.181	**36.0**

The small penalties are not small effects; they are absent ones. At 0.6 and 1.6 points against between-fold standard deviations of 0.13 to 0.18, neither is distinguishable from zero, and we read both as no measurable cross-building shift rather than as a shift that happens to be slight. The large ones are the opposite: 21.3 and 36.0 points against the same spread.

Absolute macro-F_1 is not comparable between the closed and open rows, since the open-set task has three classes and the closed-set task five, and the three retained activities are the better-populated ones. Only the penalties are comparable, and the comparison is that restricting the label space *widens* the asymmetry rather than evening it out: 0.6 against 21.3 becomes 1.6 against 36.0.

2.2. Adaptation in the open-set setting, and a metric that rewards damage

Table 2 runs the four adaptation methods of the closed-set study [1] on the open-set stream, with the rejection rule held fixed at the source-calibrated $q = 5\%$ threshold so that only the adaptation differs between rows.

In the direction with no measurable shift to recover, no method improves on the frozen source model, and the ordering is the one the closed-set study predicts from how much source normalisation each method retains. Closing the label space does not change that conclusion; it reproduces it on a harder task.

The reverse direction is where the metric becomes the finding. PETAL records the best H-score of any method, 0.2551 against source-only's 0.2276, and it does so with OS* of 0.3643 against 0.6717: it classifies the known activities roughly half as well and still wins. It also records the best UCDR in both directions.

Neither is unknown detection. PETAL's normalisation channel damages the model globally, as the closed-set study shows, which lowers its confidence on every window including the known ones. More windows fall below a fixed threshold, UCDR rises mechanically, and because H is a harmonic mean it rewards lifting whichever component was the bottleneck. Unknown detection was the bottleneck, so a method that degrades uniformly is scored as a method that detects novelty.

Table 2
Open-set TTA under the Tier-1 constraints, source-calibrated rejection at $q = 5\%$. OS* is mean per-class accuracy over known activities, UCDR the fraction of truly-unknown windows rejected, H their harmonic mean. The two degenerate rules bound the trade-off.

	$S_1 \to S_2$				$S_2 \to S_1$			
Method	OS*	UCDR	H	macro-F_1	OS*	UCDR	H	macro-F_1
Source-only	0.6639	0.3065	**0.4084**	0.5216	0.6717	0.1427	0.2276	0.5769
AdaBN	0.3818	0.2405	0.2865	0.3218	0.4058	0.1175	0.1694	0.3589
TENT	0.3826	0.2381	0.2849	0.3220	0.4065	0.1170	0.1690	0.3592
DIGA	0.5322	0.2178	0.1419	0.3856	0.6572	0.0480	0.0891	0.5450
PETAL	0.2812	**0.4375**	0.3339	0.2794	0.3643	**0.2223**	**0.2551**	0.3387
PeTTA	0.3312	0.3299	0.3208	0.2962	0.3820	0.1617	0.2148	0.3429
Reject none	0.8494	0.0000	0.0000	0.5552	0.7529	0.0000	0.0000	0.6072
Reject all	0.0000	1.0000	0.0000	0.0000	0.0000	1.0000	0.0000	0.0000

Table 3
AUROC for separating unknown from known target windows, averaged over 3 source seeds, with 95% bootstrap intervals over 2000 resamples of the target stream. 0.5 is chance. All scores are computed from the frozen source model; none uses target labels.

Score	$S_1 \to S_2$	$S_2 \to S_1$
Mahalanobis to prototype	**0.6377** [0.620, 0.655]	**0.5373** [0.522, 0.553]
Cosine to prototype	0.5668 [0.550, 0.584]	0.5070 [0.492, 0.523]
Maximum softmax	0.4952 [0.478, 0.513]	0.4434 [0.427, 0.460]
Entropy	0.4655 [0.447, 0.483]	0.4386 [0.422, 0.455]
Maximum logit	0.4580 [0.441, 0.475]	0.4343 [0.418, 0.451]
Free energy	0.4261 [0.409, 0.443]	0.4272 [0.411, 0.443]

This is worth stating as a methodological result rather than an aside. UCDR has an obvious trivial maximiser, which is why it is conventionally paired with OS*; the pairing is assumed to remove the problem. It does not. H, the standard open-set score and the one LEAD and its comparators report, is itself maximised in part by uniform degradation whenever the unknown component is the weaker of the two. An open-set result should therefore be read with the known-class number beside it, and a method that wins on H while losing substantially on OS* should be treated as a null rather than a positive. Every H-score we report below is accompanied by its two components for this reason.

2.3. Confidence-based rejection is at or below chance

Every method in Table 2 rejects on maximum softmax, which is the standard choice. Whether that score carries any information about novelty on this data can be asked directly, and independently of any threshold, by measuring how well it separates known from unknown windows in the target stream.

The four confidence-derived scores are at or below chance. Entropy, maximum logit and free energy have intervals lying entirely below 0.5 in both directions. Maximum softmax, the score the adaptation methods use, spans chance entering S_2 ([0.478, 0.513]) and lies below it entering S_1. None of the four is usefully informative, and three are reliably anti-informative, meaning the model is on average *more* confident on the activities it was never trained on than on the ones it was.

The direction of the effect is what makes it more than a null. A below-chance detector is not a weak detector; it is a detector pointing the wrong way, and a system that rejects low-confidence windows will preferentially discard the known activities it should be keeping. The two prototype-distance scores, computed in the model's representation rather than from its output, are the only ones above chance.

This parallels the per-sensor feature result of [1], where features the standard pipeline treats as its primary signal scored below stratified-random guessing across buildings. In both cases a quantity the field treats as informative is not merely uninformative on this data but carries signal of the wrong sign, and in both cases that is only visible because a chance baseline was computed.

2.4. Prototype distance helps, by about half what it first appears to

Swapping maximum softmax for the Mahalanobis distance of Table 3 improves unknown detection by a factor of 1.58 over chance entering S_2 and 1.17 entering S_1, measured at a rejection rate held equal between the two scores. That is the number we would ask a reader to carry, and it is substantially smaller than the one the natural comparison produces. We give both, because the gap between them is a general hazard in open-set reporting rather than an accident of this experiment.

Compared at the same source-calibrated threshold $q = 5\%$, the swap looks much larger: H rises from 0.4084 to 0.5176 entering S_2 and from 0.2276 to 0.4289 entering S_1, gains of 10.9 and 20.1 points. Taken at face value this is the largest effect anywhere in this study.

It is mostly an artefact of how much each score rejects. The two scores have different distributions, so a threshold placed at the same percentile of *source* validation cuts the *target* stream in very different places. Entering S_2, maximum softmax rejects 31.9% of the stream and the Mahalanobis

Table 4
UCDR with every threshold set to reject exactly 20% of the target stream. Chance is 0.2000. DIAGNOSTIC: the matched rate uses a target quantile and is not available without target data.

	$S_1 \to S_2$		$S_2 \to S_1$	
Score	UCDR	× chance	UCDR	× chance
Mahalanobis, 64-d embedding	**0.3162**	1.58	0.2342	1.17
Mahalanobis, raw 13-d input	0.3125	1.56	0.2435	1.22
Mahalanobis, 128-d	0.2988	1.49	0.2031	1.02
Mahalanobis, 256-d	0.2451	1.23	**0.2508**	1.25
Maximum softmax	0.1980	0.99	0.1975	0.99
Chance	0.2000	1.00	0.2000	1.00

distance rejects 54.0%, against a true unknown share of 40%. The higher UCDR is bought by discarding half the data, and OS* falls from 0.6639 to 0.4447 to pay for it. This is the same mechanism as PETAL's apparent win in Section 2.2, reached by a different route, and it is why that section argues for reading H beside its components.

The fair comparison fixes the rejection rate and asks which score catches more true unknowns. Table 4 sets every threshold so that exactly 20% of the target stream is rejected. Doing so requires a target quantile, so these are diagnostic figures and not Tier-1 results; they are reported to size the detector, not to be deployed.

The gain is real and it is modest. Prototype distance catches between 1.17 and 1.58 times as many unknowns as rejecting at random, and maximum softmax catches exactly as many as random, to two decimal places, in both directions. The deployable configuration of Section 1.1 uses the 64-d prototype distance at $q = 5\%$ and obtains H 0.5176 and 0.4289; those are the numbers a system would see, and the 1.58 and 1.17 figures are what the detector is actually contributing inside them.

We report this sequence in full rather than only its conclusion because the first comparison we ran was the $q = 5\%$ one, and it is the comparison an open-set paper would normally present. Nothing in the protocol flags it as wrong. It took measuring the rejection rate, which neither H nor UCDR exposes, to see that two scores held at the same nominal threshold were operating 22 percentage points apart.

2.5. A source-calibrated threshold does not transfer

The Tier-1 constraints require the rejection threshold to be fixed from source data, since calibrating it on the target stream would use the labels the setting assumes are absent. Section 2.4 showed that two scores held at the same nominal percentile rejected very different fractions of the target stream. Table 5 shows that this is systematic across the whole grid.

A threshold placed to reject one per cent of source validation rejects between 7.3% and 62.8% of the target stream, the range depending only on which score it is placed on. At $q = 5\%$ the spread is 14.8% to 71.2%. The nominal percentile, which is the quantity an open-set method reports and tunes, carries almost no information about how aggressively the system will actually reject once it is installed in another building.

The cause is the one running through this study. A threshold is a statistic of the source distribution, and source statistics do not transfer across buildings. This is the property responsible for two separate findings in the closed-set study [1]: that replacing batch-norm statistics with the target stream's is harmful, and that per-sensor features are anti-informative across buildings. It arrives here at a third place, the rejection rule. There is nothing specific to our detector in it: any open-set or universal method that fixes a threshold on source data inherits the problem, and any that fixes one on target data has left the test-time setting.

We note the practical consequence without being able to solve it. A deployment cannot observe its own rejection rate against ground truth, but it *can* observe the rate itself, which is label-free. A rejection rate far from the designer's intent is detectable in the field even when correctness is not, and we suggest reporting it alongside H as a matter of course. Calibrating a threshold that holds across domains, rather than merely across resamples of one domain, we leave open.

2.6. Depth does not help, and the detector is leaving signal unused

The prototypes of Section 2.4 are built in the model's final 64-dimensional embedding, which is trained to separate three known activities and therefore to discard whatever does not serve that. Whether novelty survives the compression is testable at no cost, by building the same prototypes at three other depths, including none at all.

The raw input features match the final embedding, 0.6392 against 0.6377 in one direction and 0.5253 against 0.5373 in the other, differences well inside the seed spread, and the intermediate blocks are worse than both. Passing the window through a trained network neither adds novelty information nor destroys it, so the rejection component of Section 1.1 does not require the classifier at all and could run beside a model of any architecture.

It is tempting to read that as a limit of the features: if a representation trained on the known classes carries no more novelty signal than the untrained input, perhaps thirteen window statistics simply do not encode what separates an unseen activity. We tested that reading rather than asserting it, and it is wrong.

All thirteen marginals differ between the withheld and retained activities, in both buildings, after correcting for the thirteen tests. A supervised classifier given the same features reaches AUROC 0.7354 and 0.6327. The information is present, and our label-free detector reaches about ten AUROC points below what a labelled one extracts from the identical description.

The single most informative feature is worth noting. In S_2 it is `t_time_cos`, the cosine of time of day, at 0.6972 on its own, which is higher than our entire multivariate detector.

Table 5
Fraction of the *target* stream rejected by a threshold set to reject q% of the *source* validation split. True unknown share is 40% entering S_2 and 30% entering S_1.

	$S_1 \to S_2$					$S_2 \to S_1$				
Score	q=1	2.5	5	10	20	q=1	2.5	5	10	20
Maximum softmax	22.9	26.5	31.9	37.7	50.5	7.3	10.5	14.8	21.7	36.5
Mahalanobis 64-d	43.6	49.0	54.0	60.0	68.6	22.1	25.8	31.0	34.9	48.9
Mahalanobis 128-d	62.8	66.5	71.2	75.6	80.5	39.6	47.3	54.1	60.5	68.8
Mahalanobis 256-d	41.4	54.2	61.8	72.2	86.9	49.6	52.7	55.9	62.4	67.4
Mahalanobis raw 13-d	54.2	58.4	69.5	76.1	87.8	34.1	37.7	42.6	56.2	64.7

Table 6
AUROC for unknown-versus-known by the representation the prototype distance is computed in, with the supervised reference of Table 7 for scale. Chance is 0.5.

Representation	$S_1 \to S_2$	$S_2 \to S_1$
Raw 13-d input, no network	**0.6392**	0.5253
Final embedding, 64-d	0.6377	**0.5373**
Second block, 128-d	0.6198	0.5076
First block, 256-d	0.5982	0.5107
Maximum softmax	0.4952	0.4434
Supervised reference (diagnostic)	0.7354	0.6327

Table 7
Is novelty present in the 13-d description? Per-feature AUROC separating unknown from known windows, and a supervised known-versus-unknown classifier trained on the target building under GroupKFold over recording day. The classifier uses target labels and is a DIAGNOSTIC upper bound, not a method.

	S_2	S_1
Features differing after Benjamini-Hochberg at 5%	13/13	13/13
Strongest single feature	0.6972	0.3700
which one	`t_time_cos`	`g_max_active`
Supervised reference, logistic	0.6620	0.5701
Supervised reference, gradient boosting	**0.7354**	**0.6327**
Our label-free detector (Table 6)	0.6377	0.5373

The withheld activities are toileting and cleaning, and they occur at different hours than meal preparation, television and music. A Mahalanobis distance to class prototypes treats all thirteen dimensions as one geometry and dilutes that; it does not exploit the one axis that most separates the groups.

So the honest statement has two parts rather than one. The learned representation adds nothing over the raw features, which is the clean negative result the depth sweep establishes. But the features are not the binding constraint: a labelled detector on the same thirteen numbers does roughly ten points better, so there is headroom a better label-free score could reach, and the most obvious place to look for it is temporal structure rather than a distance in feature space. Section 2.7 follows that up. The labelled figures should be read as a reference point rather than as a ceiling. They are what one particular supervised learner extracts, and Section 2.7 reaches 0.6616 entering S_1 with an unsupervised distance on four features, against the 0.6327 obtained there with gradient boosting on all thirteen and target labels. A bound that an unsupervised method beats was not a bound; it bounds that learner on that feature set, and the true limit of this description of the data is higher than either number. We keep the comparison because the direction of the gap is informative — a labelled detector does better than ours on the identical features, so signal is being left unused — while noting that its size is not a reliable estimate of how much.

2.7. A better detector does not produce a better system

Section 2.6 located roughly ten AUROC points of unused signal and pointed at time of day. Acting on that raises a question of method before one of performance: we identified the temporal axis by looking at target labels, so a detector built on it is selected with target information through the analyst rather than through the code. Scoring it as Tier-1 would be the error we document in others, arrived at by a different route.

The test is whether a rule that sees only source data would have chosen the same axis. Hold out each known activity from the source prototypes in turn, score that activity's own *source* windows as though they were novel, and record which dimensions detect them; averaging over the three known classes gives per-dimension weights that use no target data at all. The rule selects the temporal axis in both directions. `t_time_sin` and `t_time_cos` take the largest weights, and the four temporal dimensions average 1.58 and 1.96 against 0.74 and 0.57 for the nine activity statistics. The axis is therefore selectable without target labels on this data, which is what distinguishes a usable rule from an observation we happened to make with the target labels in front of us.

The scope of that claim needs stating plainly, because it is the strongest interpretive step in this paper. What we have shown is that one source-only rule selects the same axis that target labels identify, in two transfer directions between two apartments of one dataset, for one choice of withheld activities. Whether leave-one-class-out weighting recovers the discriminative axis in general — in other buildings, other sensor layouts, other label splits, or where novelty is not separated by time of day — is untested here and does not follow from these two agreements. The rule could equally be selecting temporal features because cyclical time is the one part of this feature vector not derived from sensor activations, in which case it would transfer, or because toileting and cleaning happen at distinctive hours in these

Table 8
Detectors that isolate or upweight the temporal axis. Δ is a paired bootstrap over the target stream against the 13-d mean baseline, 2000 resamples. Variants marked [ph] select the axis using target knowledge: they are DIAGNOSTIC upper bounds in the sense of Section 1.1, not deployable results. [t1] variants use source data only. H and the rejection rate are at the fixed $q = 5\%$ source-calibrated threshold.

	$S_1 \to S_2$			$S_2 \to S_1$		
Detector	AUROC	Δ	H / rej	AUROC	Δ	H / rej
13-d mean [t1]	0.6392	—	0.4957 / 70%	0.5253	—	0.4279 / 43%
13-d max-dim [t1]	0.6308	−0.009	**0.5235** / 57%	0.5745	+0.049	0.4410 / 54%
LOCO-weighted [t1]	0.6373	ns	0.4069 / 74%	0.5888	+0.064	**0.4825** / 55%
Temporal 4-d [ph]	0.6389	ns	0.4685 / 23%	**0.6616**	+0.136	0.4986 / 25%
Time-of-day 2-d [ph]	0.5915	−0.047	0.5053 / 26%	0.6266	+0.101	0.4207 / 16%

two particular homes, in which case it would not. We cannot distinguish those with two buildings, and a reader should treat the design rule as a hypothesis with source-only support rather than as an established method.

As a detector the temporal axis works, and it works only where there is a shift to detect. Entering S_1, every variant separates from the baseline: the source-only weighting gains 0.064 AUROC and the post-hoc temporal distance 0.136. Entering S_2, where the cross-building penalty is 1.6 points, nothing improves and two variants are significantly worse. This is the same asymmetry the adaptation methods followed in Section 2.2, now in a component that does not adapt anything.

As a system it changes nothing, and that is the more useful result. At the fixed source-calibrated threshold the source-only weighting rejects 74% of the forward stream and its H falls to 0.4069, below the 0.5176 of the configuration in Section 1.1. The better-separating score is the worse-calibrated one. The 13-d max-dim variant shows the converse: it records the best H in both directions while its AUROC is *significantly worse* in one of them, so its apparent advantage is a rejection-rate effect of the kind Section 2.4 dissects rather than better detection. We decline to recommend it for that reason.

What this establishes is a constraint rather than a method. Detection quality and deployed open-set performance come apart here: a score can gain 0.064 AUROC and lose 0.11 H at the threshold a deployment would actually use. Until a threshold can be calibrated so that it transfers, improving the detector does not improve the system, and the limiting factor in Table 5 is not the one anybody is working on.

2.8. Where the reference implementation leaves the test-time setting

The open-set and universal domain adaptation methods we build on are distributed with implementations that assume offline access to the target domain. That assumption is not stated as a limitation, and it is not visible from the papers, but it decides what the reported numbers mean. Table 9 records the four places where the reference implementation of LEAD [2] departs from the Tier-1 definition of Section 1.1, with the line that does it.

None of the four is a defect of the method as described; each is a property of the evaluation harness it ships with. The distinction matters because the first and third in particular make the reported figures an upper bound that a streaming deployment cannot approach, and because a reimplementation that honoured the constraints would not be comparable with the published numbers. All open-set results in Section 2.2 are therefore reported against our own Tier-1 source-only baseline rather than against published figures obtained under this harness.

3. Discussion

3.1. Discussion

Adding a label-space shift to a geographic one does not give adaptation more to work with. The cross-building penalty is 1.6 points entering S_2 and 36.0 entering S_1 once classes are withheld, a wider asymmetry than the closed-set 0.6 and 21.3, and in neither direction does any of the four methods improve on a frozen source model once H is read beside its components. The combined shift is harder, not richer.

What does help is replacing the rejection score, and only modestly: between 1.17 and 1.58 times chance at a matched rejection rate, against a maximum softmax baseline that sits exactly at chance, with a further 0.064 AUROC available from a source-only temporal weighting in the direction that has headroom (Section 2.7). This is a consistent reading alongside the closed-set study [1], where DIGA's prototype classifier was the one component that helped without touching normalisation statistics, but we did not isolate that component in the open-set setting and the two results are not a replication of one another. What they have in common is the direction of the leverage: in both, the useful part was a distance in representation space, and the parts that manipulate source statistics were harmful or inert.

Three things we would want a reader to take from this section are not about our system.

Metrics first. UCDR has a trivial maximiser and is conventionally paired with OS* to remove it. That pairing is insufficient, because H is itself partly maximised by uniform degradation whenever unknown detection is the weaker component, which it usually is. PETAL records the best H in

Table 9
The reference LEAD implementation (`ispc-lab/LEAD`) against the Tier-1 test-time constraints. Line numbers are from `train_target.py` unless marked. Together these make the published protocol source-free domain adaptation rather than test-time adaptation: it requires the whole target set in advance, revisits it fifty times, and evaluates on the data it adapted on.

Mechanism	Code	Constraint violated	Consequence
Multi-epoch offline training on the target set	`epochs=50` (`config/model_config.py`:15); loop at :375	Adaptation must be a single online pass over the stream	The whole target set must exist before adaptation starts, and every sample is revisited fifty times. No deployed system has this.
Adaptation and evaluation share one dataset	:360–362, both `DataLoader`s wrap the same `target_dataset`	Evaluation must not be adapted on	Every sample scored at :383 was trained on at :377. The reported score is training performance, and no held-out target split exists anywhere in the file.
Pseudo-labels built from a full pass over the evaluation set	:209 calls `obtain_LEAD_pseudo_labels` on `test_dataloader` at the start of every epoch	Predictions must depend only on samples already seen	The model sees the entire evaluation set, including its own future, before adapting. Known- and unknown-space bases are fitted on data not yet encountered.
Stream order discarded	`shuffle=True` on the adaptation loader, :360	The target stream must be processed in acquisition order	Temporal structure is destroyed and each batch mixes the whole recording period, so batch statistics are those of the full target distribution rather than of the moment.

one direction while classifying the known activities half as well as doing nothing. Open-set results should be reported with both components visible.

Thresholds second, and this is where we would put the field's effort. A threshold set to reject 1% of source validation rejects between 7% and 63% of a target stream, depending only on which score carries it. Any open-set method that calibrates on source inherits this, and any that calibrates on target has left the test-time setting.

We can now say more than that it is a nuisance: it is the binding constraint, and we have measured it as one rather than argued it. Section 2.7 improves unknown detection by a source-only rule that gains 0.064 AUROC entering S_1, closing most of the gap identified in Section 2.6. At the threshold a deployment would actually use, the same detector *loses* 0.11 H in the other direction, because separating better and calibrating better are different properties and only the first improved. The converse appears in the same table: the variant with the best H in both directions has significantly worse AUROC in one, its advantage being a rejection-rate effect rather than detection at all.

Detection quality and deployed open-set performance therefore come apart on this data, and until the threshold transfers, work on the detector does not reach the system. That makes cross-domain threshold calibration the first problem to solve rather than one of several, and it is not where the current literature, ours included until now, has been looking. A modest and concrete step is available immediately: the rejection rate is computable without labels, so a deployment can observe that it is rejecting 63% of its input when the designer intended 1%, even though it cannot observe that it is wrong. We suggest reporting it beside H as a matter of course.

Chance baselines third. The two results in this section that are most likely to generalise are both negative, and neither is visible without a chance baseline: four standard confidence scores are at or below chance as novelty detectors, and a learned representation carries no more novelty signal than the untrained input. Neither would appear as a problem in a table of H-scores alone.

The limits are the same as the closed-set study's. Two buildings, one dataset, one choice of withheld activities. The asymmetry we report is a property of this pair, and the specific values of 1.6 and 36.0 belong to it. What we would expect to transfer is the procedure: measure the penalty per cell against a target-trained model at matched sample size before running any method, report rejection rates beside H, and compute a chance baseline for any novelty score before believing it.

4. Conclusion

Adding a label-space shift to a geographic one makes cross-building activity recognition harder without giving adaptation more to work with. Across a 2×2 grid of transfer direction and label space, the methods track the measured penalty rather than the task: they help only where there is a real shift, and in the direction where the penalty is 1.6 points none of them improves on doing nothing.

What survives is small and does not adapt. A frozen source model with a Mahalanobis rejection rule, with prototypes and threshold fixed on source data before the target stream is read, reaches H-scores of 0.5176 and 0.4289; every component we removed to reach it — the four adaptation methods, the confidence-based rejection scores, and the learned embedding, which a distance on the raw thirteen features matches — was worth removing.

Three findings seem to us likely to outlive this dataset, and none of them is about our system.

The standard open-set score can be raised by degrading a model uniformly whenever unknown detection is the weaker of its two terms, as it was in both directions here, so it should never be reported without its known-class component.

Four standard confidence-based novelty scores are at or below chance here, which was visible only because a chance baseline was computed, and is a reason to compute one before trusting any detector in a new domain.

And a threshold calibrated on source data did not transfer in either direction here: the same nominal percentile rejects 7% or 63% of a target stream depending on the score it sits on, and improving the detector underneath did not help while that was true. We expect the mechanism to be general, since a threshold is a statistic of the source distribution, but we have measured it on two buildings.

The limits are those of the study it extends. Two apartments of one dataset, one choice of withheld activities, and a temporal axis that turns out to carry much of the novelty signal in these particular homes. The specific numbers belong to that pair. What we would expect to transfer is the procedure: measure the penalty per cell against a target-trained model at matched sample size before running any method, compute a chance baseline for any novelty score, report the rejection rate beside the H-score, and treat a method that wins on H while losing on known-class accuracy as a null. Calibrating a rejection threshold that holds across domains rather than across resamples of one is, on this evidence, the problem worth working on next.

Reproducibility

Code and every result file are available at `https://github.com/mariemkbaier/TTA-Test-Time-Adaptation`. Each table above names the script and result file that produced it, and `src/verify_numbers.py` checks every numeric claim in this manuscript against those files.

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